

latent fingerprint of yours and reproduce it in a mould—and this can be done—you can't be issued a new fingerprint. So how do you prove you are you?"

Even children are not safe from this identity theft. In April 2003, the story of Lisa Barraza's son was uncovered. Barraza's seven-year-old son is a typical youngster, except for the fact that since the age of one, he has piled up over \$100,000 in credit card debt. How can a child so apply for credit cards and spend that kind of cash? He Obviously can't, says his mother.

Barraza first learned of her son's debt problems when she found a credit card in his name. "My son's social security number was hijacked," says Barraza.

Bruce Schneier, founder and CTO of Counterpane Internet Security of Cupertino, California, and a recognised authority on security, says, "The reason identity theft is so powerful is that much of our security today is based on your identity."

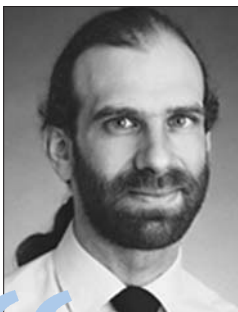
New Avenues

When data becomes increasingly centralised, things become easier for ID thieves—they can increasingly do one-stop data shopping. The cell phone is becoming a repository of the most sensitive personal data, and thus finds itself a lucrative target for attackers. Single-sign on services, such as Microsoft's Passport service, open another easy avenue for ID thieves. If a thief gets hold of one Passport ID, he or she has pretty much everything to take on the victim's identity.

Similarly, if a thief gets hold of your bank account number and online banking PIN, there's no limit to what havoc he can wreak with your finances, routing money to other accounts, sending out e-cheques, and the like.

However, many people argue along the lines that you are "more likely to be struck by lightning than becoming a victim of fraud or identity theft when they use electronic bill payment and account services". The essential thing here is to keep your password or PIN safe.

The same goes for the online use of credit cards. Although many people argue that using your credit card online is safer than using it offline because fewer people see your card number physically, the fact is that several



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unscrupulous sites will authenticate your credit card without verifying your billing address and the CVV number (the three-digit authentication number on the back of your credit card). Fortunately, such authentications are on the decline, and more and more financial institutions now require you to provide the CVV number and verify the billing address. If you're still paranoid, you can think of using services like HDFC Bank's NetSafe system wherein you are assigned a credit card number for single use, with a limit that you choose, based on your real credit card details.

You Could Be Your Own Enemy

RFID chips are particularly interesting. These chips are used in a variety of situations. For example, in healthcare, they are implanted under the skin. The intended use is to give medical practitioners access to the medical details of the person when he or she is unable to communicate. But how easy would it then be to steal such a person's identity?

Some argue that all one would need to do is to purchase an RFID scanner, scan the victim as he or she passes by, clone a new chip with the same number and implant it under their own skin. This issue—that of the stealing of information via RFID tags—may become easier as RFID readers find their way into cell phones.

In early 2004, Nokia created a reader for RFID tags for its 5140 phone. This is based on NFC (Near Field Communication) technology—which means that the phone needs to be put within an inch of a tag to be able to read it. A recent report from ABI Research says the first NFC smartphone will be out this year and goes on to predict that by 2009, more than half of all handsets will have it inbuilt.

Wireless LANs, or WLANs, are another potential area of attack. Merwyn Andrade, a contributor to the 802.11i security specification and chief technical officer at Aruba Wireless Networks, warns that ID theft could become a problem for companies using WLANs. For example, users with 802.11i wireless phones might save their passwords and PINs on the phone. "This would be like keeping your keys to your house in your wallet and then losing the wallet," he says. "Once 802.11i is pervasive, people might miss that they have to worry about authentication."



The Future Of Phishing

Dr Jonathan Tuliani, in an article floating on the Internet titled "The Future of Phishing", believes that the next few years "will see the emergence of Internet man-in-the-middle attacks". In such an attack, the user is tricked as in a usual phishing scenario, except that instead of just the user communicating with the attacker, the attacker is also communicating in real-time with the bank. "Two (or even 10) factor authentication is of no help, since the attacker does not interfere with the login process. Both the user and the bank are unaware of the presence of the attacker and believe they have a secure connection directly from one to the other.

Once established, the man-in-the-middle has complete control. He can modify instructions, even transfer funds to a different account or simply cut the user off and submit whatever instructions he desires to the bank.

"To combat this threat, it is necessary to move away from session-based security (based on a secure log-in), to message-based security (based on explicit authentication of individual transactions)."

Solutions

ID theft is the fastest growing crime in the US and is rising fast in India as well. As credit cards and online transactions, amongst other things, become more commonplace in India—banks in India are fast catching on to credit history reporting and the PAN card number may soon become what the Social Security number is in the US—it is only natural that identity thieves here will take up the same techniques.

Thinking of it as a worldwide problem, what is the answer? It probably lies in the further centralisation of data and the padlocking of that data using a mix of several techniques, including, possibly, biometrics and secure tokens.

Biometrics refers to identifying one by something one is, rather than something one has or